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Generating Cancelable Biometric Templates Using a Projection Line

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***Abstract*—Biometric authentication systems can address the problems of genuine user verification and usability suffered by the conventional cryptosystems based on the password and token. However, as biometrics is not replaceable, its privacy has become a concern. Also, if the same biometrics is used in multiple applications, it will be vulnerable to potential cross-matching attacks. In this paper, a projection line method is proposed for cancelable fingerprint template design. The main advantage of this method is its simplicity. Because there exist many possibilities projecting minutia into the projection line controlled by the whole range of possible projection line properties, the proposed approach offers non-invertible transformation and also possess good revocability property.**

***Index Terms*—security, privacy, encryption, secure template, biometrics, cancelable template.**

I. Introduction

Security and privacy have become important issues in the reliable authentication process. In most cases, password or token may be secure enough to use, however, it may not be able to distinguish between real authorized and unauthorized users who are possessing the token or password, due to its shareable property. As an alternative, biometrics (e.g. face, voice, fingerprint and iris) has been widely used to replace password or token. What is more, biometrics is relatively unique. Due to its convenience, fingerprints, which are the focus of this paper, have been popular to use [9].

On the other hand, fingerprints-based authentication has some shortcomings. First, biometric fingerprint data capture is instable since biometric reading depends on several factors, such as the device and environment. It is well known that each fingerprint imprint captured tends to be different. This will make authentication a challenging task. Secondly, biometric data stored in a database is vulnerable to attack. As fingerprint biometrics is permanently associated with a particular person and its availability is limited, once it is compromised, it will be lost forever.

It is possible to encrypt the stored biometric template, however, it still needs to decrypt it whenever the authentication is performed. As a result, the template will be exposed as often as the authentication occurred [4]. It means that biometric data is not protected anymore. Therefore, the conventional encryption/decryption process may not work well. On the other hand, performing authentication in the cipher domain is hard. This is because cryptographic and hashing algorithms, such as RSA or MD5, will produce totally different outputs although the inputs are very close.

For these reasons, it needs to construct a scheme which only the transformed (cipher) data is stored in the database where the authentication can be performed. Different from conventional cryptographic and hash functions, two relatively close inputs must produce the same outputs while not relatively close inputs must produce totally different outputs. In this research, we apply this concept on the fingerprint modality by introducing a different biometric representation.

The rest of the paper is as follows. Section 2 provides the related works. Section 3 contains the proposed scheme. Section 4 is the analysis of the experimental results. The conclusion of the research is presented in section 5.

II. Related Works

The concept of cancelable biometrics is introduced by Ratha, Connell and Bolle [13]. Here, the biometric data is transformed using an irreversible function and only the transformation and transformed biometrics are stored in the database. At the verification stage, the same transformation is performed against the biometric query before comparing it with the stored template. By using only the stored information, the original biometric template is computationally hard to reconstruct. The transformation can be carried out either in the feature or signal domain while revocation is done by generating a new transformed biometric data using a different function or parameters/keys. In this approach, there is a trade-off between non-invertability and discriminability [4]. This is because, the features of before and after transformation of the same users should be similar while different users should not. On the other hand, the transform must be irreversible. Amongst three transformation models provided in [12], [11], functional transformation has the best performance.

Furthermore, there are some properties [9], [7] which should be met in designing a biometric template protection scheme. These properties can be summarized as follows:

- Diversity. There is no cross-matching across database. Templates generated from the same biometric data using different key must not match.

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- Revocability. Revoking and reissuing a new secure (transformed) template from the same biometric information.
- Security. It is infeasible to find the original biometric information from the secure template.
- Performance. Performance of transformed template, which is often measured by using FAR, FRR and EER, is usually is not as good as non-transformed one (without protection). However, this decreasing performance should not be too much.

A research in [1] constructs an irreversible transform function by reflecting a half space of the fingerprint image into another half space and the algorithm in [5] is used for matching, afterwards. A line $ax + b$ which crosses the core point is used as the reflector. Accurately detecting the location of core point, however, is very hard [12]. The angle of the line is parameterized so it can be changed whenever it needs to generate a new template. Using this minutiae reflection, about half number of minutiae points are transformed; as a result, it is hard to construct the original minutiae position. However, after the transformation, it is possible that there will be two minutiae points whose positions are very close, which is unlikely occurred in the original template. This may result in the information leakage. Moreover, about half number of minutiae points is still in the original position (not transformed) which may also leak the biometric information.

Different from the previous researches, [8] proposes asymmetric biometric representation which requires several samples to be processed as a template. Here, a biometric sample S is divided into n parts such that $S = (s_1, ..., s_n), s_i \in \mathbb{R}$. In the enrollment process, each user provides m samples, and from these, the weights $(w_1, ..., w_n)$ are computed. Based on weights, the components are divided into q groups by considering that each group has an approximately the same weight. At last, the template representation is constructed by combining the components in each group, such that $G = (G_1, ..., G_q)$ where G_i is the component's indices in the ith group. In the matching process, there is only one sample needed for verification.

A research in [15] implements the fuzzy vault scheme [6] for the fingerprint. In this design, the cryptographic key can be recovered if only two unordered sets of minutiae points have at least k of N elements matched. Another scheme, [17], employs both local and global features of fingerprint, where a circle with a certain radius length is drawn whose center is on the core point. Two points are connected by a line and mapped to the circle in the perpendicular direction, afterwards. This is an improvement of the Sutcu's scheme [14] (cited in [17]). Further progress and issues of the fingerprint template protection are provided in [3].

III. Proposed Scheme

Motivated by [14] and [17] approaches, a projection line method is proposed for cancelable fingerprint template design. The main advantage of this method is its simplicity. There are some properties of the projection line which are parameterized, as a result, there will be many possibilities of the projecting points.

The minutiae points are extracted from fingerprint images that we can denote it as:

$$\left.\begin{array}{l} B_u \in \psi \\ B_u = \{(m_i)_u\}_{i=1}^{n} \\ m_i = (x, y, \theta)_i \end{array}\right\} \quad (1)$$

where ψ is a fingerprint domain space; B_u is a set of minutiae points extracted from a fingerprint image belonging to user u; m_i is a minutiae point number i^{th}; (x, y) is the minutiae point coordinate in the Cartesian space; θ is the minutiae orientation and n is the total number of minutiae points in B_u.

In general, our proposed scheme projects the set of minutiae points B_u into a projection line which consists of ϵ partitions and the matching decision is performed based on the vector generated from these partitions. This approach can be divided into four parts, namely: alignment, projection, quantization and matching. A simple example is presented in Fig. 1, where $n = 4$ and $\epsilon = 8$.

In detail, the proposed scheme can be described as follows.

A. Alignment

The biometric query is aligned with the template by plotting its minutiae points into a Cartesian coordinate system. The core points of both fingerprint template and query are positioned in the same coordinate and orientation by performing translation and rotation, such that their core points' coordinate and orientation are overlap. The core point m_0 is set as the center of the space and its orientation is to be $x - axis$. Hence, both the template's and query's core point will be $m_0(0, 0)$ whose orientation is 0°. An example of this process is shown in Fig. 2. In practice, plotting of fingerprints template and query into a Cartesian system is done separately as their images are not stored in the database.

B. Projection

The minutiae points $\{m_i\}_{i=1}^{n}$ are projected to the line: $y = \rho x + c$, where c is the y-intercepts of the line and ρ is the slope of the line, such that:

$$\arctan(\rho) = \alpha \quad (2)$$

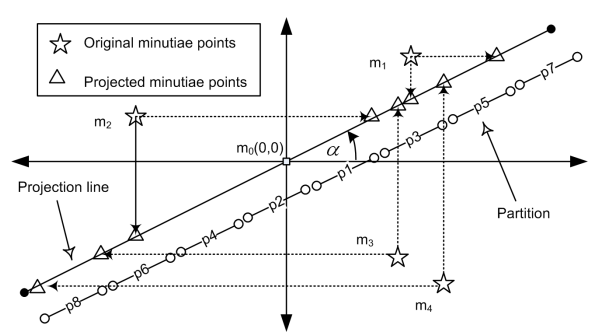

Fig. 1. Minutiae point's projection scheme

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where $\alpha \in [0, \pi], \alpha \neq \frac{\pi}{2}$. As this line crosses the axes in $(0, 0)$, the constant c is zero. We set the degree of the slope α fixed to a certain value; however, it can also be parameterized to vary the template.

We define two different methods in projecting the minutiae points as depicted in Fig. 3.

- (x, y)-projection: two coordinate properties, (x, y), are projected to the line, such that:

$$m_i \Rightarrow \{(m_r)_i, (m_r)_{n+i}\} \tag{3}$$

  where $(m_r)_i$ and $(m_r)_{n+i}$ are m_i projection parallel to the x and y axis, respectively. So, the total number of projection points is twice of the minutiae points', as illustrated in Fig. 3a, where the number of original and projection minutiae points are 2 and 4, respectively
- (x, y, θ)-projection: the projection includes three properties of (x, y, θ), such that:

$$m_i \Rightarrow \{(m_r)_i, (m_r)_{n+i}, (m_r)_{2n+i}\} \tag{4}$$

  where $(m_r)_{2n+i}$ is m_i projection parallel to the θ while $(m_r)_i$ and $(m_r)_{n+i}$ are same as previous definition in (3). Here, the total number of points in the projection line will be three times greater than minutiae points', as illustrated in Fig. 3b, where the number of original and projection minutiae points are 2 and 6, respectively.

Let C be all possible point coordinates in a Cartesian domain space, those original and projection points can be denoted as: $\{m_i, (m_r)_i, (m_r)_{n+i}, (m_r)_{2n+i}\} \subset C$. The set of projection points $(B_r)_u$ is:

$$\left.\begin{array}{l} (B_r)_u = \{(m_r)_i\}_{i=1}^{l} \\ (m_r)_i = (x_r, y_r)_i \end{array}\right\} \tag{5}$$

where $l = 2n$ or $l = 3n$, depends on the projection method being used.

C. Quantization

In general, quantization is carried out by dividing the projection line into some partitions and counting the number of points in each partition, afterwards [14], [17]. This generates a vector which is used in the matching step. In our approach, the vector is determined by two factors: projection line position and partition.

1) Projection Line Position: As the output of the previous steps, the projection points will spread over the line. Not all of those points, however, are used in the quantization process for some reasons (discussed in IV-A). Only points within the line's length, L, are included in the matching process. To determine the line position, which actually also determines the points to be processed, we calculate the center of the line, $m_c(x_c, y_c)$, such that $\frac{L}{2}$ is located in m_c.

We construct two different approaches in determining the position of $m_c(x_c, y_c)$ in the coordinate space. The first approach is by locating the center of the line fixed in the core point $m_0(0, 0)$ - we call it as center (ctr)-approach, such that:

$$\left.\begin{array}{l} x_c = 0 \\ y_c = 0 \end{array}\right\} \tag{6}$$

The second approach is by locating it relative to the template's projection points - we call it as relative (rel)-approach. Here, x_c is set in the middle of the maximum and minimum value of projection points' abscissa. Let $\Pi x_r = \{(x_r)_i\}_{i=1}^{l}$ be the set of all projection points' abscissa in B_r. Center of the projection line is defined as:

$$\left.\begin{array}{l} x_c = \frac{max(\Pi x_r) - min(\Pi x_r)}{2} + min(\Pi x_r) \\ y_c = \rho x_c \end{array}\right\} \tag{7}$$

2) Partition: The projection line is divided into some partitions whose characteristics depend on the key:

$$\kappa = \{\epsilon, \mu_1, \sigma\} \tag{8}$$

where ϵ, μ_1 and σ are the total number of partitions, the length of partitions and the indices of partitions, respectively, such that: $L = \epsilon * \mu_1$. If the projection line is mapped to the $x-axis$, the actual length of each partition will be $\mu_2 = \mu_1 * \cos(\alpha)$, as depicted in Fig. 4.

The quantization step will generate a vector $v_u = [s_1 s_2 ... s_\epsilon]$, where v_u is a vector of user u; s_i is the total number of projection points in the partition i^{th}, $1 \leq i \leq \epsilon$; $(1, 2, ...\epsilon)$ are the indices of partitions (σ). This vector will be a template which is stored in the database. A simple example is shown in Fig. 5, where both ctr and rel approaches are applied, $\epsilon = 6$ and $\sigma = (6, 4, 2, 1, 3, 5)$. The first approach will generate v_u = [3 1 2 3 0 2] while the second will generate v_u = [1 3 3 2 2 1]. Not only generating different vectors, those approaches

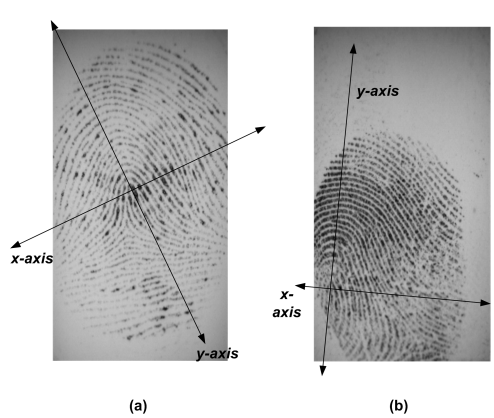

Fig. 2. Alignment between template and query (a) template (b) query

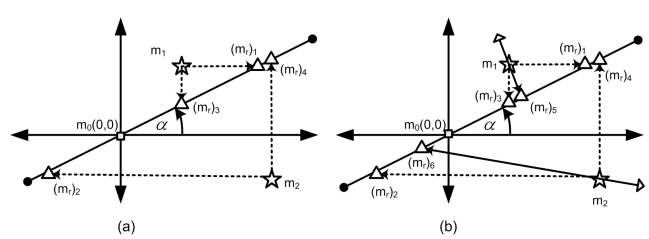

Fig. 3. Minutiae projection approaches (a) (x, y)-projection; (b) (x, y, θ)-projection

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also result in different numbers of projection points covered by the line, namely 11 and 12 for ctr and rel approaches, respectively.

D. Matching

Queries to be matched to the template will also be projected and quantized same as the template in the previous steps. Matching is performed by comparing the value of the vector template with the query. Let $dis(v_u, v'_u)$ denotes the distance between the vector template v_u and query v'_u. If the template and query are relatively close, then the query is judged as matched to the template and otherwise. Let τ_1 and τ_2 be specified thresholds for the v_u and v'_u to be close. We apply two different definitions of *distance* for measuring the closeness between v_u and v'_u, namely:

- We apply the mean absolute error (MAE), such that:

$$dis(v_u, v'_u) = \frac{1}{\epsilon}\Sigma_{i=1}^{\epsilon}|(s_i)_u - (s_i)'_u| \tag{9}$$

  If $dis(v_u, v'_u) \leq \tau_1$ then v'_u is matched to v_u
- We perform direct comparison (DC) between $(s_i)_u$ and $(s_i)'_u$, such that:

$$(s_i)_u - \delta \leq (s_i)'_u \leq (s_i)_u + \delta \tag{10}$$

  If there are at least τ_2 $(s_i)'_u$ meet this criterion, then v'_u is matched to v_u. For the experiment, we set $\delta = 1$.

IV. Experimental Results

For the experiment, we use a public database FVC2002Db2a [2] that contains 8 databases. We use 100 images of them, which consists of 50 images each of the first two databases. Genuine test is performed by comparing corresponding images of the second to the first databases while impostor test is by comparing each image of the second to all images in the first databases except its corresponding image. Our previous work FOMFE model [16] has been used for singular point detection as FOMFE model does not require any knowledge about the raw fingerprint image. The evaluation result is represented in a Receiver Operating Characteristic (ROC) curve and Equal Error Rate (EER) for measuring the False Acceptance Rate (FAR) and False Rejection Rate (FRR). We use Verifinger software [10] for extracting the fingerprint features.

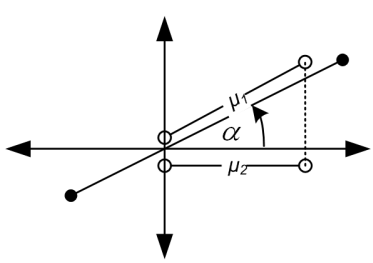

Fig. 4. Partition's length based on the x-axis

A. Performance

It is observed that the rel-approach (7) which is applied to the (x, y)-projection coordinates (3) and evaluated using MAE (9) gives the best performance as shown in ROC curves in Fig. 6 whose parameters used for testing and its error rate are provided in table I and Fig. 7, respectively. One of factors which affects this error rate is the quality of singular point detection.

In this experiment, the rel-approach is better than the ctr-approach because it adds dependency to the two points (maximum and minimum points) in the template. Furthermore, each template is likely to have different maximum and minimum points, so the projection line covers more unique area; consequently, the number of projection points being counted will also be more specific. However, this approach may not work well if these two points are missing or moving too far due to limitation in the fingerprint reading process. In addition, the use of the minutiae orientation in the projection process does not increase the performance. Overall, the result relies on the accuracy of the core point detection which is harder to achieve than minutiae detection.

Number of minutiae point. Due to some factors, such as device and environment, the number of minutiae points read by the device is instable. Therefore, a pair of template and query which is extracted from an identical fingerprint may have a different number of minutiae points. In the quantization step (section III-C), there are about 95% of projection points $((m_r)_i)$ for both template and query covered by the projection line. The other 5% is not used for some reasons. For example, its location is too far from $m_c(x_c, y_c)$ which can cause an increase of FAR because the line contains

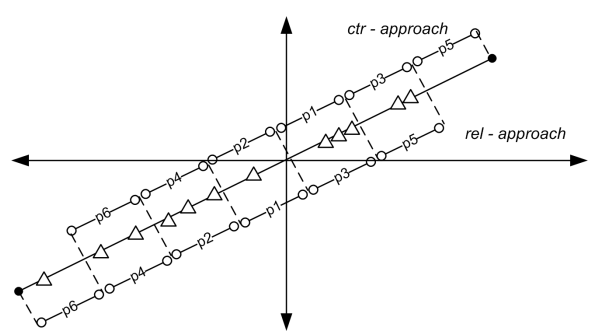

Fig. 5. Different partition position in the projection line

TABLE I
Parameters used in the experiments

Combination approaches	μ_1	α
$rel - xy - mae$	38	27
$rel - xy - dc$	37	30
$rel - xy\theta - mae$	31	30
$rel - xy\theta - dc$	30	30
$ctr - xy - mae$	26	30
$ctr - xy - dc$	28	30
$ctr - xy\theta - mae$	22	30
$ctr - xy\theta - dc$	18	30

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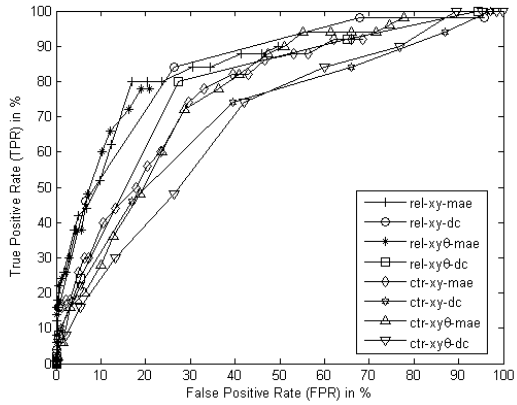

Fig. 6. ROC curves of the experiment results

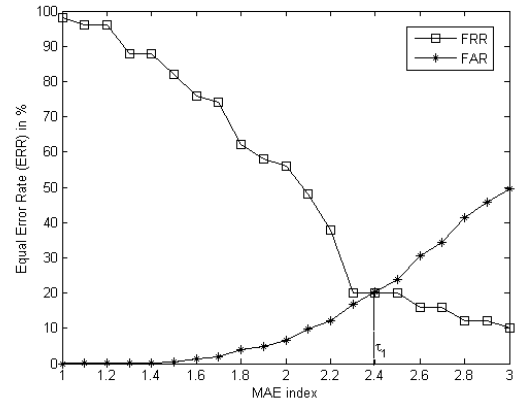

Fig. 7. Equal Error Rate (EER) of the $rel - xy - mae$ approach

empty partitions, as depicted in Fig. 8.

In addition, the difference of total projection point numbers being counted for the matching process (i.e. covered by the projection line) between template and query may relate to the MAE index, regardless of the threshold being used. This difference affects the total number of the vector's elements. Furthermore, the number of total projection points l itself, for both template and query, depends on the number of original points n which has been read by the device. Fig. 9 shows that in general, an increase of point differences will make the MAE index higher. Consequently, the possibility of queries to be rejected is also higher.

B. Revocability and Diversity

Revoking and reissuing a new vector template are carried out by generating a new vector template using a different key, $\kappa = \{\epsilon, \mu_1, \sigma\}$. The old and new templates must have a certain distance such that those two are relatively different and do not correlate each other. It means that it should meet the diversity requirement.

For example, let $\kappa' = \{\epsilon', \mu_1', \sigma'\}$ be the old key to generate the revoking template v_u', and $\kappa = \{\epsilon, \mu_1, \sigma\}$ be the new key for the new template v_u. Let $\epsilon' = \epsilon = 8$ and $\mu_1' = \mu_1$, $\sigma' = (\epsilon', (\epsilon' - 2), (\epsilon' - 4), ..., 2, 1, 3, ..., (\epsilon' - 5), (\epsilon' - 3)(\epsilon' - 1))$ and $\sigma = (\epsilon, (\epsilon - 1), (\epsilon - 2), ..., 2, 1)$, such that $\sigma' = (8, 6, 4, 2, 1, 3, 5, 7)$ and $\sigma = (8, 7, 6, 5, 4, 3, 2, 1)$. Matching is done in two steps. First, we compare each pair of images which is taken from the same finger. The result shows that that all queries in the second database are rejected by their corresponding template in the first database. Secondly, non-identical fingerprints which are from different fingers are tested. The result shows that about 98% of them are rejected. This is still higher than when $\sigma' = \sigma$ which is around 80%. So, in term of diversity, the true rejection rate is higher, which is good. It means that in this case, around 98% of v_u meets the specified distance.

As an alternative, a new vector template can also be issued by parameterizing either the slope (ρ) or constant (c) of the projection line; or varying the length of partitions, such that $\mu_1 = \{(\mu_r)_i\}_{i=1}^{\epsilon}$, where $(\mu_r)_i \in \mathbb{R}$ and $\Sigma_{i=1}^{\epsilon}(\mu_r)_i = L$; instead of varying the key κ.

C. Security

In constructing vector template v_u, the indices of the vector elements (partitions), σ, are parameterized which also depends on the total number of partitions, ϵ; so, there will be $\epsilon!$ partition order (σ) possibilities. For example, for a projection line with 11 partitions (intervals), there are 39916800 possible ways of partition. Therefore, increasing the total number of partitions will increase the security.

In case v_u is compromised, B_u is still not revealed because v_u does not contain the coordinate values of B_u. In the worst case that both v_u and κ are compromised, the maximum information that can be revealed is the upper and lower bounds of $(m_r)_i$ coordinate. On the other hand, to reconstruct m_i, it needs the exact $(m_r)_i$ and its partner coordinates, $(m_r)_{n+i}$, $(m_r)_{2n+i}$ (3, 4). Therefore, it is still infeasible to reconstruct B_u given both v_u and κ.

V. Conclusion

Biometrics offer many advantages to use in the authentication process and might replace password and token in the future. However, its security and privacy have been an issue as once it is compromised, the effect will be forever. In this paper, a biometric protection template design scheme is proposed.

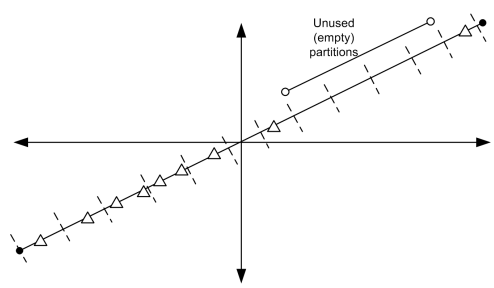

Fig. 8. A projection line which contains empty partitions

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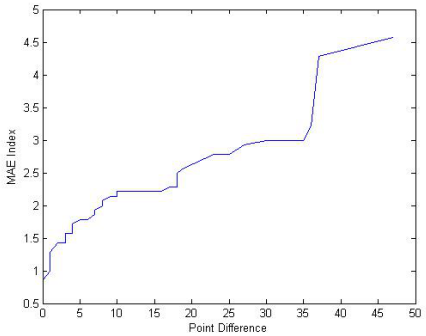

Fig. 9. Correlation between MAE index and number of point difference between template and query in the projection line

The idea is to project minutiae points into a line in the Cartesian coordinate and divide the line into some partitions. In this scheme, the representation of biometric templates is different from the original biometric features.

The proposed scheme has been evaluated using public database FVC2002Db2a in term of performance, revocability, diversity and security. Analysis of the experiment results is also presented. Similar to many existing cancelable fingerprint template design schemes, the proposed projection line cancelable template design also requires good quality of singular point detection as the feature transformation much relies on it. Investigation of singular point independent cancelable template schemes is our future research topic.

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